

Glossary: Generative AI Engineering and Fine-Tuning Transformers

Welcome! This alphabetized glossary contains many terms used in this course. Understanding these terms is essential when working in the industry, participating in user groups, and participating in other certificate programs.

Estimated reading time: 5 minutes

Glossary

Term	Definition
AdamW optimizer	A stochastic optimization method that modifies the implementation of weight decay separately from the gradient update.
Additive fine-tuning	A method that involves adding new task-specific layers or components to a pre-trained model.
AG News dataset	A subdataset of the AG corpus consisting of news articles used for text classification tasks.
Agent autonomy	The capability of an AI agent to operate independently, make decisions, and perform actions without continuous human intervention.
Agent memory	A mechanism that allows AI agents to store, retrieve, and use past interactions or information over time.
Agent orchestration	The coordination and management of multiple AI agents or tools to work together toward a common goal.
Agent planning	The process by which an AI agent decomposes a high-level goal into actionable steps.
Agent tool use	The ability of an AI agent to invoke external tools, APIs, or services to complete tasks.
AI agent	An autonomous system that perceives its environment, reasons about tasks, and takes actions to achieve defined goals.
Autonomous agent	An AI agent capable of operating independently without continuous human intervention.
Bidirectional and Auto-Regressive Transformers (BART)	A sequence-to-sequence large language model that uses an encoder-decoder architecture for contextual understanding and text generation.
Bidirectional Representation of Transformers (BERT)	An open-source model that provides deeply bidirectional, unsupervised language representations pretrained on a plain text corpus.
Bidirectional Representation of Transformers (BERT) tokenizer	A component that processes and converts input text into tokens suitable for the BERT model.
Bradley-Terry model	A probability model used to predict the outcome of pairwise comparisons between items or entities.
Chain-of-thought (CoT)	An AI reasoning technique that breaks complex problems into intermediate logical steps.
Chat model	A model designed for conversational interaction that understands prompts and responds in a human-like manner.
ChatGPT	An artificial intelligence chatbot developed by OpenAI.
Context encoder	A neural network architecture commonly used for image inpainting tasks.
Contextual embeddings	Embeddings that represent words based on their surrounding context within a sequence.
Data leakage	A situation where sensitive or private information is unintentionally exposed or used improperly.
Data loader	A utility in machine learning frameworks that loads and batches data for training or inference.
Datapoint	An individual, identifiable element within a dataset.
Decision-making policy	A strategy used by AI agents to select actions based on goals, observations, and constraints.
Dense Passage Retrieval (DPR)	A retrieval method that fetches relevant text passages using similarity between dense vector representations of queries and documents.
Direct preference optimization (DPO)	A fine-tuning method that aligns LLMs with human preferences without reinforcement learning or reward models.
Embedding store (Vector store)	A database that stores vector embeddings and enables similarity search, commonly used in RAG pipelines.
Facebook AI Similarity Search (Faiss)	A library developed by Meta for efficient similarity search over high-dimensional vectors.
Faiss index	A data structure that enables fast similarity search over vector embeddings.
Few-shot prompt	A prompting technique where a small number of examples are provided to guide model behavior.
Fine-tuning	A supervised process that adapts a pre-trained model to specific downstream tasks.
Generative Pre-trained Transformer (GPT)	A self-supervised model trained to predict the next token in a sequence using a decoder-only architecture.
GitHub	A platform used to host, manage, and collaborate on software development projects.
Global Vectors for Word Representation (GloVe) dataset	An unsupervised learning algorithm that generates word embeddings based on word co-occurrence statistics.
Graphic Processing Unit (GPU)	Specialized hardware designed to accelerate graphics rendering and parallel computation.
Hugging Face	A platform providing open-source tools, libraries, and pretrained models for machine learning and NLP.
IMDb dataset	A dataset containing movie reviews and metadata, commonly used for sentiment analysis tasks.
In-context learning	A technique where task instructions or examples are embedded directly within the input prompt.

Term	Definition
Knowledge base	A structured or unstructured collection of documents, data, or embeddings used by a RAG system.
LangChain	An open-source framework that simplifies building applications using large language models.
LangChain chains	Sequences of calls that connect prompts, models, and tools together.
LangChain community	A collection of third-party integrations built on top of LangChain Core interfaces.
LangChain-Core	The foundational module of LangChain that provides base abstractions and expression language.
Language model	A model that predicts or generates text based on previously seen tokens.
Large language models (LLMs)	Foundation models trained on vast datasets with many parameters to generate and understand natural language.
LLaMA	A family of large language models developed by Meta AI for efficient and scalable language understanding.
Low-rank adaptation (LoRA)	A parameter-efficient fine-tuning technique that injects low-rank matrices into model layers.
Low-rank transformations	Techniques that approximate large matrices using smaller ones to improve computational efficiency.
Machine learning	A field of AI focused on enabling systems to learn patterns and make decisions from data.
Model inference	The process of using a trained machine learning model to make predictions on new data.
Multi-agent system	A system consisting of multiple interacting AI agents that collaborate or compete to solve tasks.
Natural language processing (NLP)	A subfield of AI focused on enabling computers to understand, interpret, and generate human language.
Neural network	A computational model inspired by biological neural systems, consisting of layers of interconnected nodes.
Parameter-efficient fine-tuning (PEFT)	Techniques that adapt large pre-trained models using fewer trainable parameters.
Prefix tuning	A fine-tuning approach that prepends learnable embeddings to transformer attention mechanisms.
Prompt engineering	The practice of designing effective prompts to guide AI model outputs.
Prompt injection	A technique that inserts task-specific tokens or instructions into model inputs.
Prompt template	A predefined structure used to generate consistent prompts dynamically.
Prompt tuning	A method that learns continuous prompt embeddings prepended to input text.
P-tuning	A prompt-based fine-tuning approach where learnable embeddings are optimized during training.
Python	A high-level programming language widely used in data science and AI development.
PyTorch	An open-source deep learning framework that provides dynamic computation graphs and tensor operations.
PyTorch tensors	The core data structure in PyTorch used to represent multi-dimensional arrays.
PyTorch tokenizer	A tool that converts raw text into tokens compatible with PyTorch-based models.
Quantized Low-Rank Adaptation (QLoRA)	A method that combines quantization with LoRA to reduce memory usage and computation cost.
RAG (Retrieval-Augmented Generation)	An AI approach that combines information retrieval with text generation, enabling language models to produce responses grounded in external knowledge.
Reactive agent	An AI agent that responds to immediate environmental stimuli without long-term planning or memory.
Re-ranking	A process that reorders retrieved documents based on relevance scores to improve the quality of context.
Reinforcement learning from human feedback (RLHF)	A fine-tuning approach that uses human feedback to improve model alignment and performance.
Reparameterization-based methods	Techniques that reparameterize model weights using low-rank structures for efficiency.
Retriever	A component in a RAG system responsible for fetching relevant documents or passages.
Retrieval-augmented generation (RAG)	A framework that enhances large language model outputs by retrieving relevant external knowledge during generation.
Scoring function	A function used to evaluate or measure the quality of predictions.
Selective fine-tuning	A fine-tuning approach that updates only specific layers or parameters of a model.
Self-consistency	A technique that improves reasoning reliability by aggregating multiple reasoning paths.
Soft prompts	Learnable prompt representations that guide model behavior without modifying model weights.
Supervised fine-tuning (SFT)	A transfer learning technique where labeled data is used to adapt a pre-trained model.
Supervised fine-tuning (SFT) trainer	A training framework used to apply supervised fine-tuning to pre-trained models.
Task decomposition	The process of breaking down complex goals into smaller, manageable tasks.
Text classifier	A machine learning model that assigns predefined categories to text inputs.
Tokenization	The process of converting raw text into discrete tokens for model processing.
Tool-calling agent	An AI agent designed to determine when and how to use external tools or APIs.
Transformer model	A neural network architecture based on self-attention, widely used for NLP tasks.
Vector	A mathematical representation of numerical values commonly used in machine learning.

Term	Definition
Vector averaging	The process of computing the mean vector from a set of vectors.
watsonx.ai	An IBM platform providing access to a suite of enterprise-grade large language models.
WatsonxLLM	A wrapper interface for IBM watsonx.ai foundation models.
Weight-decomposed low-rank adaptation (DoRA)	A LoRA variant that adjusts rank based on component magnitude to improve efficiency.
Zero-shot prompt	A prompting technique where a model performs a task without seeing prior examples.



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